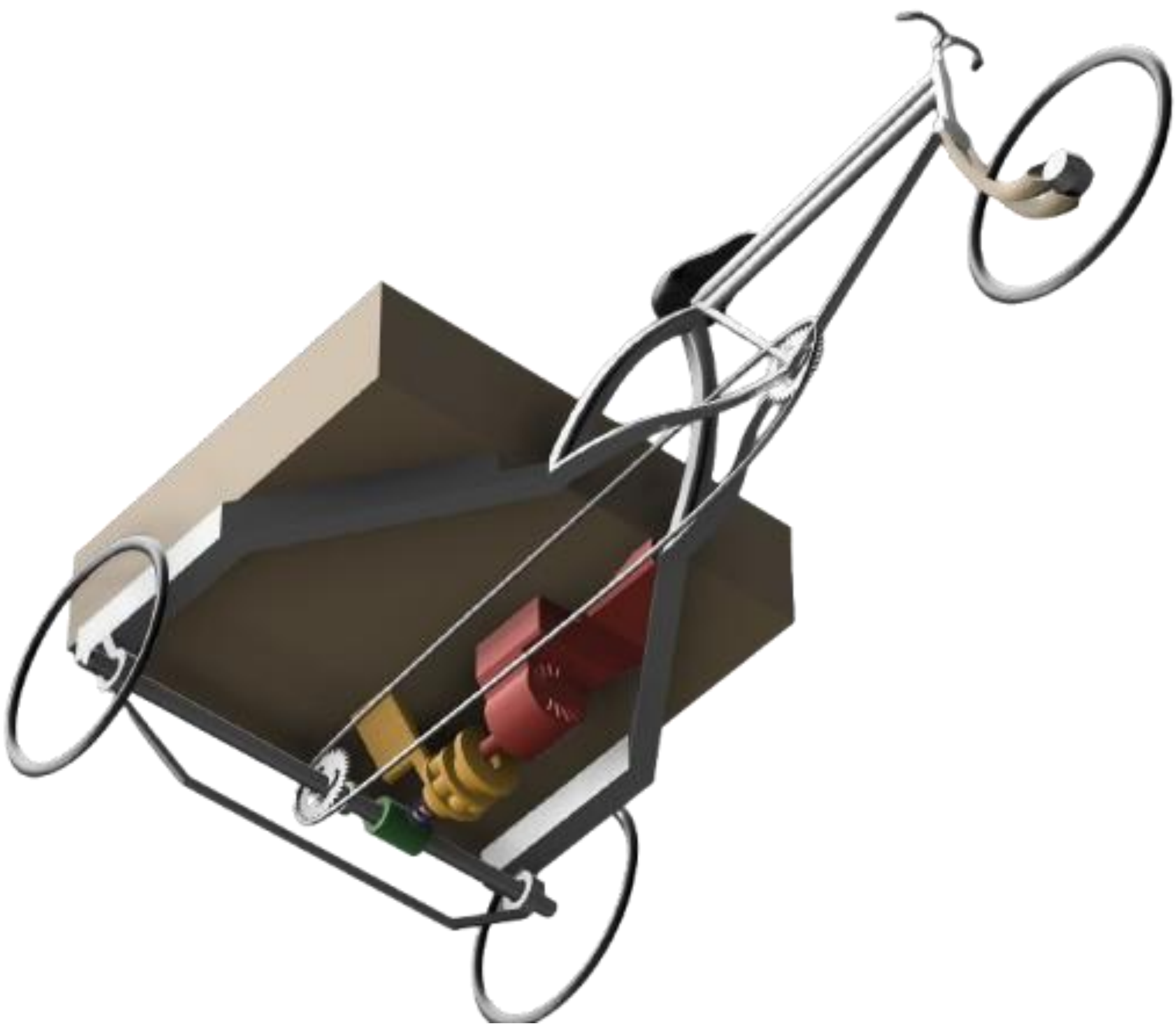
TORQUE-ASSIST INJECTION ARCHITECTURE FOR SOLID-AXLE CARGO TRICYCLES UNDER INFORMAL-USE CONSTRAINTS

Volume 1 / Issue 1 / 2026

Anurag Mishra

Independent Engineering Research

New Delhi, India
Open engineering preprint



Constraint-driven electric assist for informal logistics vehicles

Abstract

Solid-axle cargo tricycles are widely used in informal urban logistics across low-resource contexts, where human operators routinely transport loads exceeding ergonomic limits. While electric assistance has been explored in formal cargo platforms, most existing solutions are incompatible with the mechanical, economic, and regulatory constraints of informal-use tricycles, which typically lack freewheels, standardized axles, or provisions for drivetrain modification. This work presents a torque-assist injection architecture designed specifically for solid-axle cargo tricycles operating under informal-use constraints.

The system is defined by strict non-negotiable boundaries: no axle machining or replacement, no electronic speed control, full reversibility, passive back-drivability, and mechanical failure modes that default to unassisted human propulsion. Rather than pursuing performance optimization, the architecture prioritizes safety, maintainability, and compatibility with heterogeneous vehicle geometries. The contribution of this paper is a constraint-driven system architecture that injects assistive torque downstream of human input while preserving direct mechanical control by the operator at all times.

This paper documents the system layout, component logic, safety boundaries, and failure-mode behavior, and proposes a validation framework appropriate for informal-use deployment. No claims are made regarding regulatory certification or production readiness. The objective is to establish a publicly inspectable engineering record of a low-complexity torque-assist architecture suitable for further testing, critique, and iteration.

Section 1: Introduction

Human-powered cargo tricycles play a critical role in last-mile logistics across many low- and middle-income regions. In dense urban environments, these vehicles enable goods movement in areas where motorized transport is economically inaccessible, spatially impractical, or legally restricted. Operators routinely transport heavy payloads for extended durations under demanding environmental and traffic conditions, often relying entirely on human power.

Despite their importance, traditional cargo tricycles remain largely absent from formal vehicle design and electrification research. Most engineering work on electric cargo transport has focused on purpose-built electric cargo bicycles, electric rickshaws, or light electric vehicles designed for formal markets. These platforms typically assume standardized geometries, integrated electrical systems, replaceable drivetrains, and access to structured maintenance networks. In contrast, cargo tricycles used in informal logistics are characterized by heterogeneous geometries, solid rear axles without differentials, minimal onboard electrical systems, and dependence on local workshop repair ecosystems [A representative example of the target vehicle class is shown in Fig. 1.]

Figure 1; Reference vehicle class.
A typical solid-axle human-powered cargo tricycle used in informal urban logistics. Vehicles of this class generally employ a rigid rear axle without a differential, directly driven by human power via chain and sprocket.

Electrifying these vehicles therefore presents a distinct engineering challenge. Conventional electrification approaches; such as hub motors, drivetrain replacement, or speed-controlled propulsion; conflict with the mechanical, economic, and operational realities of informal-use tricycles. Modifications that require axle machining, specialized servicing, high upfront capital, or complex electronic systems are unlikely to be adopted and may introduce failure modes that threaten operator livelihoods.

The objective of this work is not full vehicle electrification, but fatigue-reducing torque assistance compatible with existing cargo tricycles and their operating ecosystem. The design goal is a retrofit architecture that preserves manual operability, avoids irreversible modification of the rear axle, maintains compatibility with informal repair practices, and ensures that electrical or mechanical failures default to unassisted human propulsion.

This paper presents a constraint-driven system architecture that injects assistive electric torque into the existing solid rear axle of cargo tricycles without replacing the drivetrain or electronically commanding vehicle speed [The high-level concept of the proposed retrofit architecture is illustrated in Fig. 2.] Rather than optimizing for performance or efficiency, the proposed architecture prioritizes safety, reversibility, and deployability in informal-use contexts.

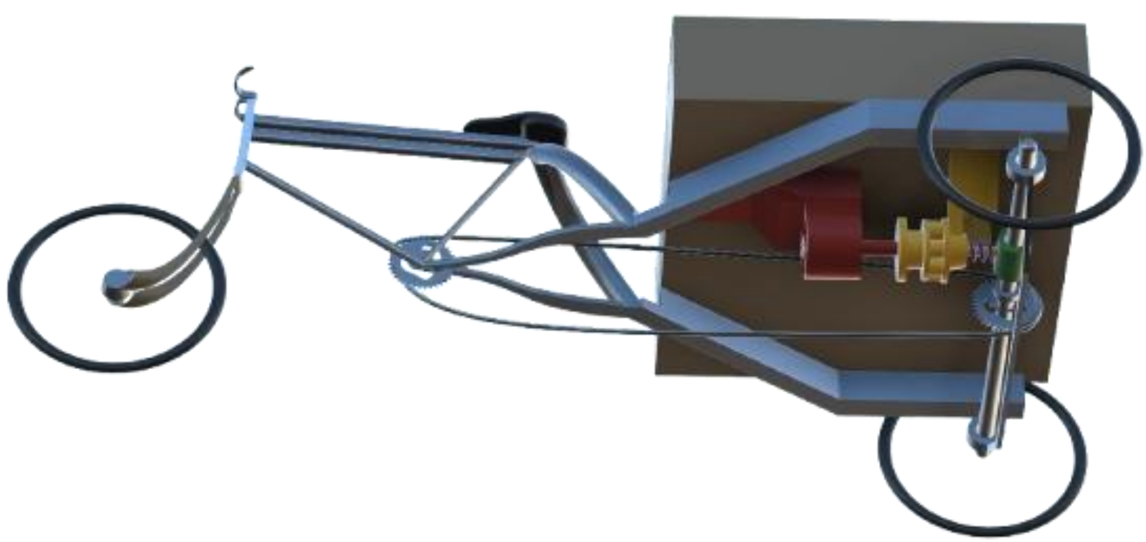


Figure 2;Conceptual system layout (CAD)
Simplified CAD representation of the proposed torque-assist injection system, showing motor, reduction stage, and non-invasive axle coupling. Exact geometries and mounting vary across vehicle instances. Supplementary materials and CAD models are available at: https://tirare-webpage.github.io/#cad

The primary contribution of this work is the formalization of a torque-injection retrofit architecture for solid-axle cargo tricycles, together with a proposed methodology for validation and early-stage deployment. This contribution aims to establish a foundation for further experimental study and iterative development of electric-assist solutions tailored to informal logistics systems.

2. Related Work

Electrically assisted micromobility has received substantial attention in recent years, particularly in the context of e-bikes, cargo bicycles, and light electric vehicles. However, the majority of existing research and commercial development focuses on purpose-built platforms designed for formal markets and standardized manufacturing environments.

2.1 Electric Bicycles and Cargo Bikes

Extensive research exists on pedal-assist electric bicycles (pedelecs) and electric cargo bicycles. Prior work typically emphasizes energy efficiency, battery management, motor control strategies, and rider–motor interaction models. Modern pedelec systems commonly rely on torque sensors, cadence sensors, and closed-loop speed or power control to deliver regulated assistance.

Commercial cargo e-bikes similarly integrate hub motors or mid-drive systems into purpose-designed frames. These platforms assume standardized bottom brackets, replaceable drivetrains, integrated wiring harnesses, and access to manufacturer-supported servicing. Safety and regulatory compliance in these systems often depend on electronic speed limiting, diagnostic systems, and firmware-controlled operation.

While highly effective in formal markets, these approaches are tightly coupled to new vehicle manufacturing and do not readily translate to legacy cargo tricycles with non-standard geometries and minimal onboard electronics.

2.2 Electrification of Informal Transport Vehicles

Research on electrification in low- and middle-income regions has primarily focused on electric two-wheelers, electric rickshaws, and small electric utility vehicles. These studies typically examine battery swapping ecosystems, cost optimization, charging infrastructure, and environmental impact.

Electric rickshaws, in particular, have achieved widespread adoption. However, these vehicles are usually designed as electric-first platforms or undergo full drivetrain replacement. The engineering assumption in most such systems is that propulsion will be motor-dominant, with human power either absent or secondary.

Human-powered cargo tricycles used in informal logistics differ fundamentally from electric rickshaws. They are pedal-driven, mechanically simple, and maintained by small local workshops. Full electrification approaches can introduce cost, complexity, and repair barriers that conflict with existing usage patterns and economic constraints.

2.3 Retrofit and Appropriate Technology Approaches

A smaller body of work explores retrofit electrification and appropriate technology solutions for mobility and logistics. These approaches emphasize affordability, maintainability, and compatibility with local repair ecosystems. However, most retrofit kits still assume drivetrain modification, wheel replacement, or hub motor installation.

Hub motor retrofits are particularly common due to their installation simplicity. Yet hub motors are incompatible with solid-axle cargo tricycles lacking differentials, where wheel replacement is non-trivial and axle modifications risk structural integrity. Additionally, many retrofit systems rely on speed-controlled electronic propulsion, introducing failure modes that may be unsuitable for informal-use contexts.

There remains limited formal documentation of architectures specifically designed to inject assistive torque into existing solid axles while preserving full manual rideability and avoiding irreversible mechanical modification.

2.4 Research Gap

Current literature provides strong foundations for electric bicycles, electric rickshaws, and light electric vehicles, but does not directly address the electrification of legacy solid-axle cargo tricycles operating in informal logistics systems.

This gap motivates the present work. The architecture proposed in this paper targets a narrow but widespread vehicle class and prioritizes constraint-driven design, safety-by-default failure modes, and compatibility with informal repair practices. The intent is to establish a documented baseline architecture suitable for further experimental validation and iterative development.

3. Operational Constraints in Informal Vehicle Retrofitting

Electrification strategies developed for formal vehicle markets cannot be directly transferred to legacy cargo tricycles used in informal logistics. These vehicles operate within heterogeneous manufacturing conditions, rely on decentralized repair ecosystems, and must remain functional under frequent mechanical wear and uncertain maintenance histories. Consequently, the proposed architecture is defined by a set of explicit operational constraints that bound the permissible design space. These constraints are treated as first-order requirements rather than secondary engineering preferences.

3.1 Target Vehicle Class

The system targets a narrow but widely deployed vehicle configuration consisting of human-powered cargo tricycles with a single front steering wheel and two rear wheels mounted on a solid rear axle without a differential. Propulsion is provided through a pedal–chain drivetrain, and the vehicle frame is assumed to be fabricated from steel.

A key characteristic of this vehicle class is the presence of a one-way mechanism (freewheel or ratchet) within the pedal drivetrain, preventing axle rotation from back-driving the pedals. This feature enables torque assistance to be introduced without compromising rider safety or pedaling ergonomics.

Vehicles outside this configuration—such as differential-equipped tricycles, electric rickshaws, shaft-driven drivetrains, aluminum or composite frames, or motor-dominant vehicles—are considered outside the scope of this work. Restricting the supported vehicle class enables a consistent safety model and integration strategy.

3.2 Rear Axle Constraints

The rear axle is the dominant structural element governing retrofit feasibility. In the target vehicle class, the axle is typically a solid steel shaft that simultaneously serves as the primary load-bearing member and torque transmission component. These axles are rarely precision-machined and are generally installed and serviced using low-precision tooling.

To preserve structural integrity and maintain compatibility with informal repair practices, the retrofit architecture is constrained to avoid any operation that alters the axle's structural or bearing interface. Specifically, the architecture must:

- Avoid axle removal or machining
- Avoid the introduction of axial loads on existing bearings
- Preserve bearing alignment and mounting geometry
- Remain installable using minimal tooling

Axles exhibiting pre-existing damage, misalignment, or excessive wear are considered unsuitable for retrofit integration.

3.3 Frame and Mounting Constraints

Frames in this vehicle class are typically constructed from mild steel and are routinely repaired through local welding and fabrication. The architecture therefore assumes that the frame can tolerate additional localized loads associated with motor reaction torque. However, integration must not compromise structural serviceability or repairability.

All mechanical integration must therefore satisfy the following requirements:

- No frame cutting or permanent structural modification
- Bolt-on or clamp-based mounting only
- Full reversibility and replaceability of mounts

These constraints preserve compatibility with existing workshop practices and ensure that the retrofit does not introduce irreversible dependencies.

3.4 Electrical System Constraints

Informal-use cargo tricycles generally lack integrated electrical systems. The retrofit is therefore designed as a self-contained low-voltage subsystem operating at or below 60 V nominal. All electrical components are independent of the base vehicle and non-essential to its operation.

This architectural separation ensures that electrical failure cannot compromise the fundamental mobility of the vehicle and prevents coupling between mechanical reliability and electrical subsystem reliability.

3.5 Control and Operational Constraints

The rider remains the sole controller of vehicle motion. The system is explicitly designed as assistive technology rather than primary propulsion. Human pedaling remains the dominant source of propulsion, and the electrical subsystem provides supplementary torque intended to reduce operator fatigue.

Accordingly, the architecture excludes:

- Electronic speed control
- Autonomous or semi-autonomous behaviors
- Complex training requirements

These constraints prioritize familiarity, user trust, and behavioral continuity.

3.6 Environmental Constraints

The system is designed for sustained outdoor operation under environmental conditions typical of urban logistics:

- Ambient temperature range: 0 °C to 45 °C
- Exposure to dust and light rain
- Daily operation up to approximately 10 hours
- No guaranteed indoor storage

Operation in flooded conditions, submerged environments, or under high-pressure washing is outside the intended operating envelope.

3.7 Safety Boundary Conditions

Several boundary conditions are treated as non-negotiable safety requirements. The architecture is considered invalid if it results in:

- Removal of the human drivetrain
- Rigid or locked coupling of the rear axle
- Speed-commanded electronic propulsion
- Elimination of mechanical slip behavior
- Transfer of battery ownership to vehicle operators

These constraints enforce a fail-safe philosophy in which the vehicle remains manually operable and mechanically controllable under all failure modes.

---

4. System Architecture

4.1 System Overview

4.1.1 System Objective

The Tirare retrofit system provides bounded electric torque assistance to the solid rear axle of an existing cargo tricycle while preserving the vehicle's native human-powered drivetrain, mechanical topology, and repair ecosystem.

The system is explicitly designed as an assistive subsystem rather than a propulsion replacement. Human pedaling remains the primary source of vehicle motion, and electrical torque is introduced only in response to deliberate rider input. At all times, the vehicle must remain fully rideable without electrical power.

The architecture is therefore defined around three invariants:

- Preservation of manual operability
- Preservation of existing drivetrain geometry
- Preservation of compatibility with informal repair practices

---

4.1.2 Functional Architecture

The system is organized around three interacting flows:

1. Human control flow
   The rider continuously provides propulsion through pedaling and requests torque assistance through a simple manual input.
2. Electrical power flow
   Energy is supplied by a removable battery and delivered to the motor through protection and control electronics.
3. Mechanical torque flow
   Motor torque is reduced and coupled externally into the existing rear axle through a non-invasive interface.

These flows operate concurrently but remain architecturally separable to preserve fail-safe behavior.

4.1.3 End-to-End Control Flow

Assistive torque is generated only in response to direct human input:

Human -> Assist Input- > Current-Limited Controller -> Motor Torque

The control architecture intentionally excludes closed-loop speed regulation or autonomous behavior. In particular, the system performs:

- No speed sensing
- No speed targeting
- No autonomous decision-making

Vehicle speed remains an emergent outcome of rider effort, terrain, and load.

4.1.4 Electrical Power Path

The electrical subsystem is arranged as a protected low-voltage power chain:

Battery -> Manual Disconnect -> Fuse -> Motor Controller -> Motor

Electrical independence from the base vehicle is a fundamental architectural principle. Any electrical failure interrupts assistive torque delivery while preserving full manual vehicle functionality.

4.1.5 Mechanical Torque Path

Assistive torque is transmitted to the vehicle through an external coupling path:

Motor -> Reduction Unit -> External Axle Coupler -> Rear Axle -> Rear Wheels

The rear axle remains the mechanically authoritative element of the drivetrain. The architecture preserves:

- Back-drivability by the rider
- Continuous human propulsion capability
- Passive mechanical behavior under fault conditions

4.1.6 Authority and Non-Authority of the System

The system does <u>not</u> control vehicle speed or motion. Instead, vehicle speed emerges from:

- Rider input
- Payload and terrain
- Passive drivetrain dynamics

This separation ensures that the retrofit cannot command motion independently of the rider.

4.1.7 Default and Failure Behavior

The architecture is defined around passive fail-safe behavior. Under all fault or non-powered conditions, the vehicle reverts to standard manual operation.

State	System Behavior
Powered, no rider input	Zero assist
Powered, assist requested	Current-limited torque assist
Battery removed	Fully manual operation
Electrical fault	Fully manual operation
Mechanical slip	Fully manual operation

No failure mode exists in which the system can:

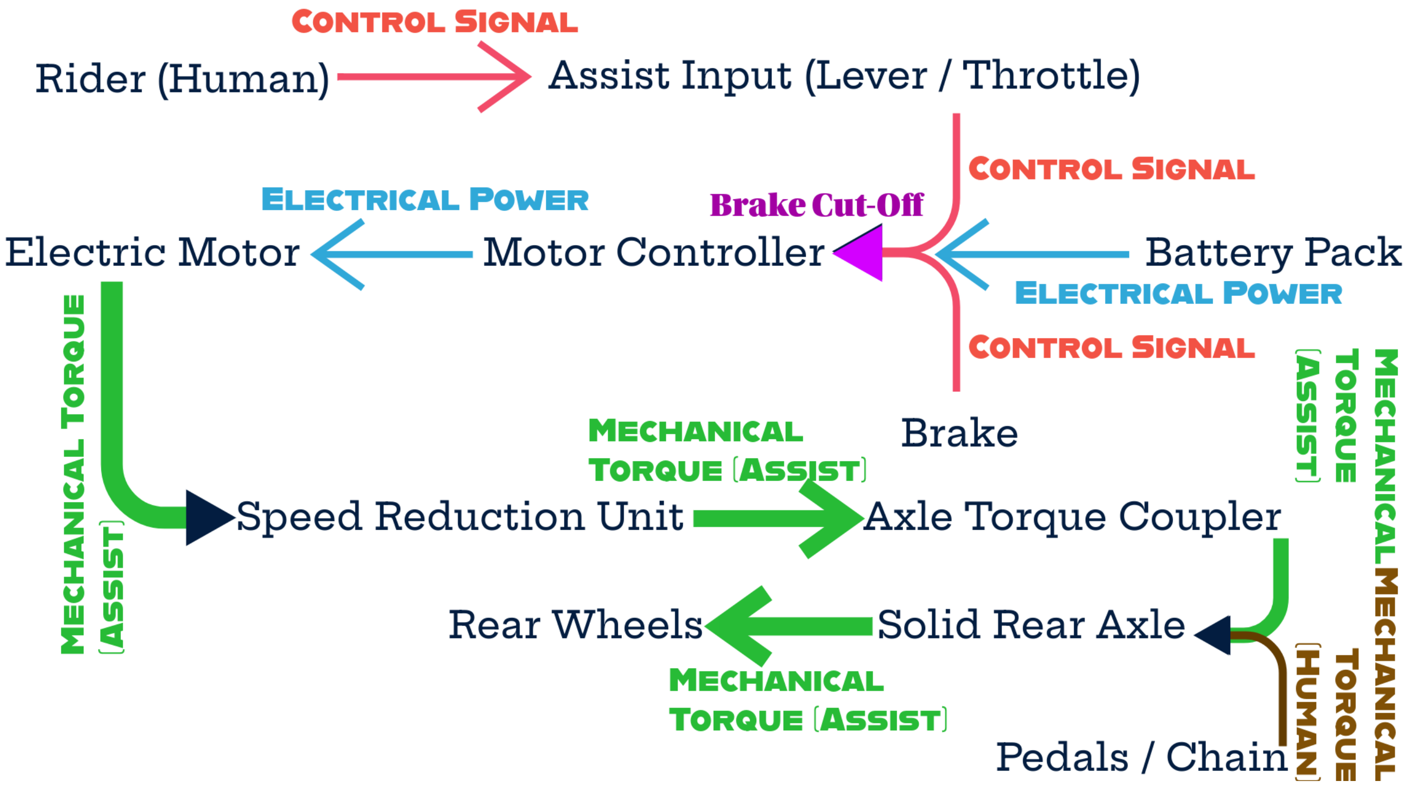
- Lock the rear axle
- Command vehicle motion
- Prevent pedaling

4.1.8 Physical Subsystem Layout

The retrofit consists of five primary physical modules:

- Frame-mounted motor–reduction assembly
- External axle coupling interface
- Removable battery pack
- Handle-mounted assist input device
- Inline safety disconnect

All modules are designed to be bolt-on, independently serviceable, and individually replaceable.



4.1.9 Architectural Diagram

Figure D2-1 illustrates the end-to-end system architecture spanning human control, electrical power delivery, and mechanical torque transmission.

4.2 Component Architecture and Subsystem Requirements

This section defines the primary hardware subsystems used to implement the architecture described in Section 4.1. Component requirements are specified as performance envelopes rather than vendor-specific selections.

4.2.1 Assist Motor

Assistive torque is generated using a frame-mounted brushless DC (BLDC) motor. The motor is intentionally decoupled from the axle and mounted to the vehicle frame in order to preserve the native drivetrain topology.

Required performance envelope:

- Nominal voltage: 36–48 V
- Continuous output power: 250–350 W
- Peak output power (≤30 s): ≥500 W
- Continuous shaft torque: ≥1.2–1.5 N·m
- High speed constant (mechanical reduction performed externally)

The motor must be fully back-drivable and capable of frequent start–stop operation. Integrated gearmotors are excluded to maintain serviceability and mechanical modularity.

Acceptable implementations include industrial BLDC motors and mid-drive e-bike motors operated without pedal-sensing subsystems. Hub motors and axle-mounted motors are explicitly excluded.

4.2.2 Mechanical Reduction Stage

Motor output is converted from high-speed/low-torque to low-speed/high-torque using an external reduction stage mounted to the frame.

Required characteristics:

- Reduction ratio: 15:1 to 30:1
- Back-drivable transmission
- Shock-tolerant operation
- Frame-mountable housing
- Output shaft suitable for external coupling (round or keyed)

A cycloidal reducer is preferred due to its shock tolerance and high reduction capability. Industrial planetary reducers are considered acceptable alternatives.

4.2.3 Axle Torque Coupling Interface

Motor torque is introduced into the existing solid axle through a non-invasive coupling interface. This element functions as the mechanical fuse of the system.

Accepted implementations include:

- Split clamp collars
- Friction sleeves
- External hub-style couplers

The coupling must:

- Require no axle machining or welding
- Introduce no axial loads to the axle
- Slip before axle or frame damage occurs
- Passively disengage under overload

Materials are typically steel or aluminum with a friction interface.

4.2.4 Motor Controller

Torque delivery is regulated by a current-controlled BLDC motor controller.

Required performance envelope:

- Nominal voltage: 36–48 V
- Continuous current: ≥20 A
- Peak current: ≥35–40 A
- Thermal derating capability

Required control features:

- Analog torque command input (0–5 V or equivalent)
- Brake cut-off input
- Overcurrent protection
- Undervoltage lockout

Speed regulation, connectivity, display integration, and mobile application support are intentionally excluded to preserve architectural simplicity.

4.2.5 Energy Storage System

Energy is provided by a removable Lithium Iron Phosphate (LFP) battery pack selected for cycle life, thermal stability, and abuse tolerance.

Nominal specifications:

- Voltage: 36 V (12s) or 48 V (16s)
- Capacity: 400–600 Wh
- Continuous discharge capability: ≥1C
- Cycle life target: ≥2000 cycles

Mechanical requirements include a rigid enclosure, carry handle, and keyed mounting interface. Electrical protection is provided by an integrated battery management system supporting over-voltage, under-voltage, and over-temperature protection.

Battery ownership is assumed to remain workshop-controlled rather than user-owned.

4.2.6 Human Assist Input Device

Assist torque is requested through a spring-return lever or thumb throttle mounted on the handlebar.

Required behavior:

- Analog voltage output
- Zero output at rest
- Normally-open behavior
- Approximately linear response

This input commands torque only, not vehicle speed.

4.2.7 Safety Subsystems

The electrical safety chain includes:

- DC-rated main fuse sized at 1.25× maximum continuous current
- Manual battery disconnect located near the battery pack
- Brake-actuated torque cut-off signal

These elements ensure immediate torque removal under fault or braking conditions.

4.2.8 Wiring and Interconnects

Conductor sizing is defined by current class:

- Battery → controller: 10–12 AWG
- Controller → motor: 12–14 AWG
- Signal wiring: 18–22 AWG

High-current connections must use anti-spark DC connectors and strain-relieved terminations. Solder-only joints are not permitted in high-current paths.

4.2.9 Structural Mounting Hardware

Subsystem mounting uses:

- Mild steel brackets
- Grade 8.8 or higher fasteners
- Lock nuts or thread-locking compounds
- Elastomer bushings for vibration isolation

4.3 Electrical System Architecture

4.3.1 Design Objectives

The electrical subsystem is designed to deliver controlled motor torque while remaining strictly non-essential to vehicle operability. The architecture enforces current-limited torque delivery, guarantees fail-safe transition to zero assist, and ensures that electrical faults cannot impede manual propulsion.

Under no condition may electrical failure:

- Lock the drivetrain
- Command vehicle motion
- Prevent pedaling

This constraint defines the electrical system as a removable assistive subsystem rather than a propulsion system.

4.3.2 Power Distribution Architecture

The high-current power path is strictly unidirectional and contains no parallel loads. Electrical energy flows through a minimal protection chain before reaching the motor controller:

Battery- > Manual Disconnect -> Main Fuse- > Motor Controller -> Three-phase Motor Output

The absence of auxiliary electrical loads simplifies fault isolation and reduces wiring complexity.

4.3.3 Low-Voltage Control Architecture

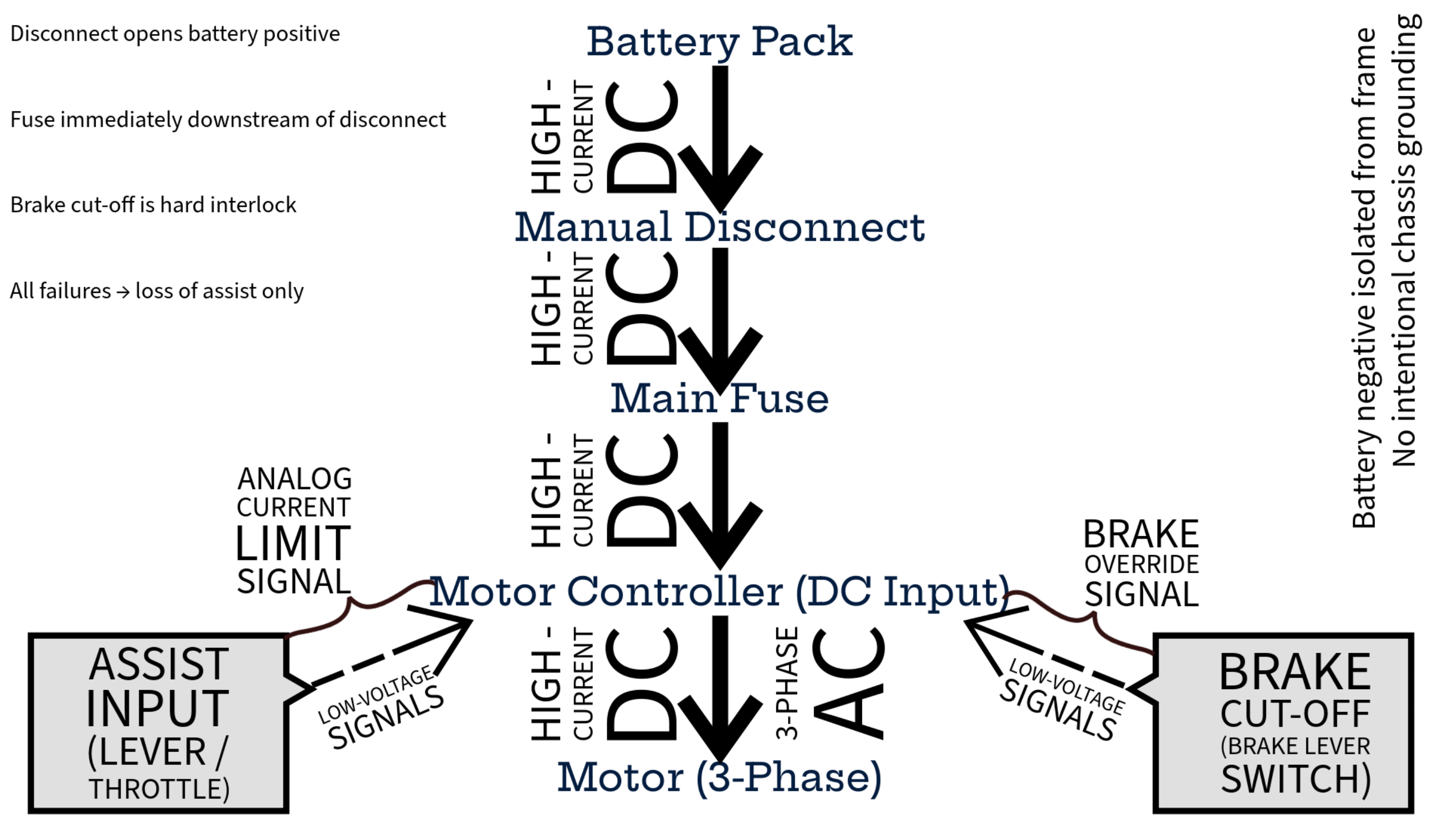
A separate low-voltage signal path communicates rider intent and safety interlocks to the controller:

Assist input device -> analog torque command input
Brake lever signal -> controller brake cut-off input

No speed sensing, closed-loop velocity control, or autonomous decision-making is present in the control path.

4.3.4 Electrical Block Diagram

Diagram D2-2 presents the high-level electrical architecture and illustrates the separation between power and control pathways.



4.3.5 Conductor Sizing and Routing

High-current conductor sizing is determined by continuous current requirements:

- Battery to disconnect: 10–12 AWG
- Disconnect to fuse: 10–12 AWG
- Fuse to controller: 10–12 AWG
- Controller to motor (three phase): 12–14 AWG

High-current conductors must be abrasion-protected, strain-relieved, and routed to avoid sharp bends.

Low-voltage and signal wiring (20–22 AWG) is routed separately from motor phase wiring to reduce electromagnetic interference and vibration-induced fatigue.

4.3.6 Safety Chain Integration

The electrical safety chain consists of three hardware elements arranged in series.

Manual disconnect.

A tool-free, clearly marked mechanical disconnect interrupts the battery positive terminal and guarantees immediate transition to manual-only operation.

Main fuse.

A DC-rated fuse located directly downstream of the disconnect provides overcurrent protection. The fuse rating is selected to exceed controller continuous current while remaining below the controller's absolute maximum rating.

Brake cut-off interlock.

A brake-actuated signal directly disables controller output without software mediation. This constitutes a hardware-level torque interlock.

4.3.7 Electrical Isolation Strategy

The battery negative terminal and controller enclosure are electrically isolated from the vehicle frame unless otherwise required by the controller manufacturer. The architecture intentionally avoids frame grounding in order to reduce corrosion risk, eliminate ground loops, and limit fault propagation.

4.3.8 Connectorization Requirements

Connector selection prioritizes user safety and serviceability:

- Battery connectors: anti-spark, polarized, and touch-safe
- Motor phase connectors: high-current rated and strain-relieved
- Signal connectors: locking and vibration-resistant

Exposed live conductors are not permitted.

4.3.9 Electrical Fault Behavior

All credible electrical faults are designed to produce a single outcome: loss of assist torque while preserving manual rideability. Fault conditions including battery removal, fuse activation, disconnect operation, control input failure, brake signal faults, or controller malfunction result in zero motor output.

The architecture therefore contains no failure mode capable of producing unintended torque.

4.4 Mechanical System Architecture

4.4.1 Mechanical Design Objectives

The mechanical subsystem is responsible for transmitting assist torque to the rear axle while preserving the native human-powered drivetrain and guaranteeing fail-safe disengagement. The retrofit must remain mechanically non-obstructive when unpowered and must be installable without permanent vehicle modification.

Accordingly, the architecture is designed to:

- Transmit limited assist torque to the drive axle
- Preserve full manual rideability
- Fail in a neutral or disengaged state
- Avoid irreversible modification of the host vehicle

4.4.2 Drive Architecture Selection

The selected topology is a <u>frame-mounted mid-drive assist system</u> coupled to the rear axle through an external reduction stage. This architecture was selected to:

- Preserve wheel interchangeability and existing maintenance practices
- Enable torque limiting through mechanical reduction
- Minimize unsprung mass
- Maintain freewheeling behavior during unpowered operation

Hub motors and axle-mounted motors are excluded due to incompatibility with informal repair ecosystems and axle serviceability.

4.4.3 Motor Mounting Architecture

The electric motor is a shaft-output brushless DC unit mounted to the vehicle frame using a bolt-on bracket. The mounting system must:

- Maintain shaft alignment under torsional loading
- Resist motor reaction torque
- Allow motor removal without axle disassembly

The bracket is constructed from ≥5 mm steel or aluminum plate and interfaces with existing frame members using clamp-based or existing bolt locations. Slotted mounting provisions enable tension adjustment for the reduction stage.

4.4.4 Reduction Stage Architecture

Motor speed is converted to axle-compatible torque using an external reduction stage. Acceptable transmission mechanisms include timing belt or chain drives.

The reduction ratio is selected to:

- Limit maximum axle torque below structural limits
- Prevent motor overspeed under no-load conditions

Direct drive coupling is explicitly excluded. All rotating elements must be guarded, and pulley or sprocket alignment tolerance is maintained within ±0.5 mm.

4.4.5 Axle Coupling and Overrunning Mechanism

Torque is injected into the axle through a freewheel or overrunning clutch. This component is critical to ensuring mechanical transparency of the system during unpowered operation.

The overrunning mechanism:

- Allows pedaling without motor drag
- Prevents motor back-driving
- Ensures zero resistance when unpowered

Rigid axle coupling is prohibited. The coupling must tolerate minor misalignment and must not introduce axial loads to axle bearings.

4.4.6 Mechanical Load Path

The assist torque path is defined as:

Motor -> Reduction Stage -> Overrunning Clutch -> Axle ->> Wheels

Assist loads are intentionally isolated from the pedal drivetrain. No assist torque passes through the crankset, chainring, or pedals, preventing unintended biomechanical loading of the rider.

4.4.7 Clearances and Packaging

Minimum mechanical clearances are enforced to prevent interference and damage:

- Rotating components to frame: ≥5 mm
- Electrical wiring to rotating elements: ≥10 mm
- Ground clearance: unchanged from baseline vehicle

4.4.8 Vibration and Acoustic Considerations

Rubber isolation elements may be used between the motor and mounting bracket to reduce vibration transmission. Fasteners must employ thread-locking compounds, and transmission tension must follow manufacturer specifications. The assembly is designed to avoid resonant excitation in the 20–60 Hz band during steady operation.

4.4.9 Mechanical Failure Behavior

The mechanical architecture is designed so that all credible failures result in loss of assist while preserving manual operability. Failure of the transmission, clutch, motor, or mounting structure leads to passive disengagement rather than drivetrain obstruction.

4.4.10 Integration Sequence

The mechanical integration sequence is defined to minimize misalignment risk and ensure safe assembly:

1. Installation of motor mounting bracket
2. Installation of motor unit
3. Installation of reduction stage
4. Installation of axle overrunning clutch
5. Alignment verification
6. Installation of protective guards
7. Manual rotation check to confirm free motion

This sequence reflects the dependency structure of the mechanical load path and ensures verification prior to powered testing.

4.5 Human Interface and Control Architecture

4.5.1 Control Philosophy

The control system is intentionally minimal and assistive. The rider remains the sole agent responsible for propulsion, speed selection, and braking. The electronic subsystem does not command velocity; it only permits or limits motor torque.

This philosophy ensures that the vehicle behaves as a conventional cargo tricycle under all conditions, with electric assist functioning strictly as a fatigue-reduction mechanism.

4.5.2 Assist Demand Input

Rider intent is communicated through a handlebar-mounted analog input device (rotary throttle or linear lever) connected directly to the motor controller's current-limit input.

The input sets the maximum permitted assist torque, not vehicle speed. At zero input, commanded motor current is zero and no additional mechanical drag is introduced.

A monotonic mapping between input position and torque limit is used:

Input Position	Assist Behavior
0%	No assist
1–30%	Low assist
30–70%	Moderate assist
70–100%	Maximum permitted assist

Unless otherwise specified, this mapping is linear.

4.5.3 Emergent Speed Regulation

Vehicle speed is not directly measured or regulated. Instead, speed emerges from the interaction of:

- Human pedaling power
- Terrain and payload
- Rolling and aerodynamic resistance
- The fixed motor power ceiling

The architecture intentionally excludes cruise control, speed-hold logic, and autonomous acceleration.

4.5.4 Pedal Dependency Model

The system does not require pedal sensors. Assist is permitted only when the rider both pedals and commands assist through the input device. If pedaling ceases, vehicle speed decreases naturally due to resistive forces, and assist torque diminishes as axle speed approaches the motor's no-load condition.

This passive dependency eliminates the need for additional sensing and simplifies the control stack.

4.5.5 Brake Override Interlock

A brake-actuated cut-off signal provides a hardware-priority torque disable. When braking is detected:

- Motor current is immediately disabled
- Assist input is ignored

The braking experience must remain indistinguishable from that of a non-assisted vehicle, and assist torque must never oppose braking.

4.5.6 Startup and Shutdown Behavior

Power-on:
Assist defaults to zero. The rider must intentionally request torque; no startup torque is permitted.

Power-off:
Assist ceases immediately. The overrunning clutch prevents motor drag, and manual pedaling continues normally.

4.5.7 Rider Feedback

The minimum feedback requirement is a power-state indicator. Additional indicators (assist level or battery state) are optional and not required for core operation. No display system is necessary.

4.5.8 Control Fail-Safe Behavior

All credible control failures result in loss of assist torque. Input disconnection, controller faults, brake signal faults, or power loss cannot produce persistent or unintended assist.

The architecture therefore ensures that assist cannot persist without active rider intent.

4.5.9 Minimal Training Requirements

System operation requires only basic instruction:

1. Begin pedaling before applying assist
2. Increase assist gradually
3. Release assist before braking
4. Power off when parking

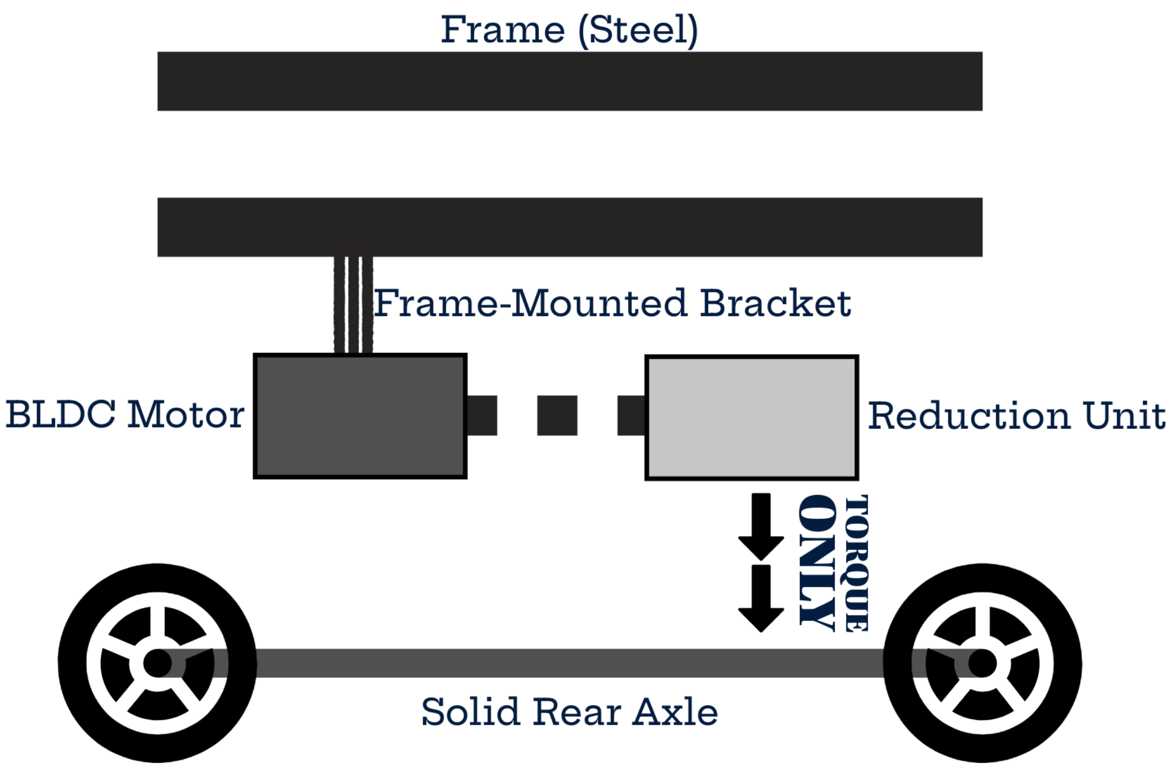
No specialized training is required for safe operation.

---

5. Prototype Implementation and Installation

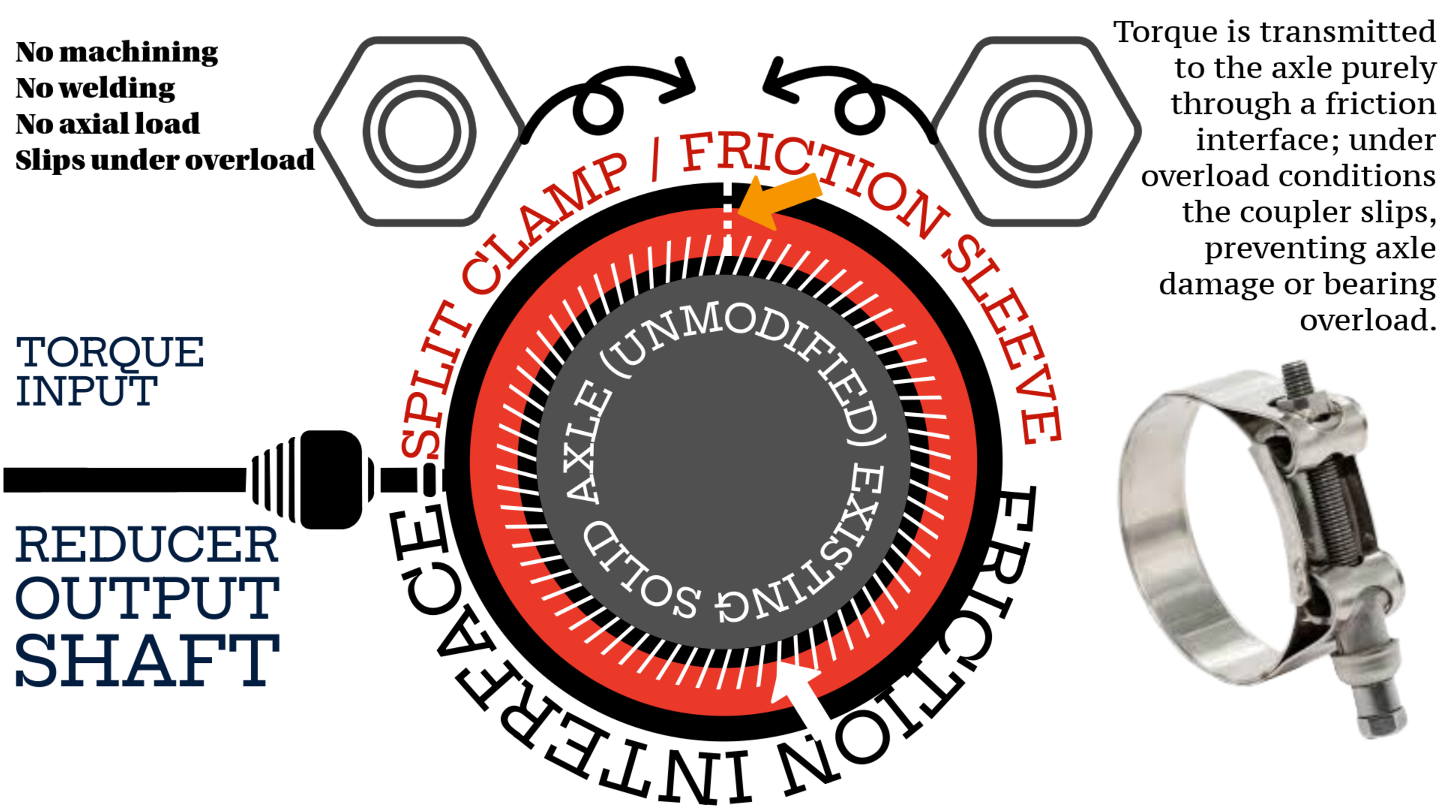
5.1 Motor and Reducer Mounting -> mechanical realization

Frame-supported motor, no axle load, real mounting constraints.



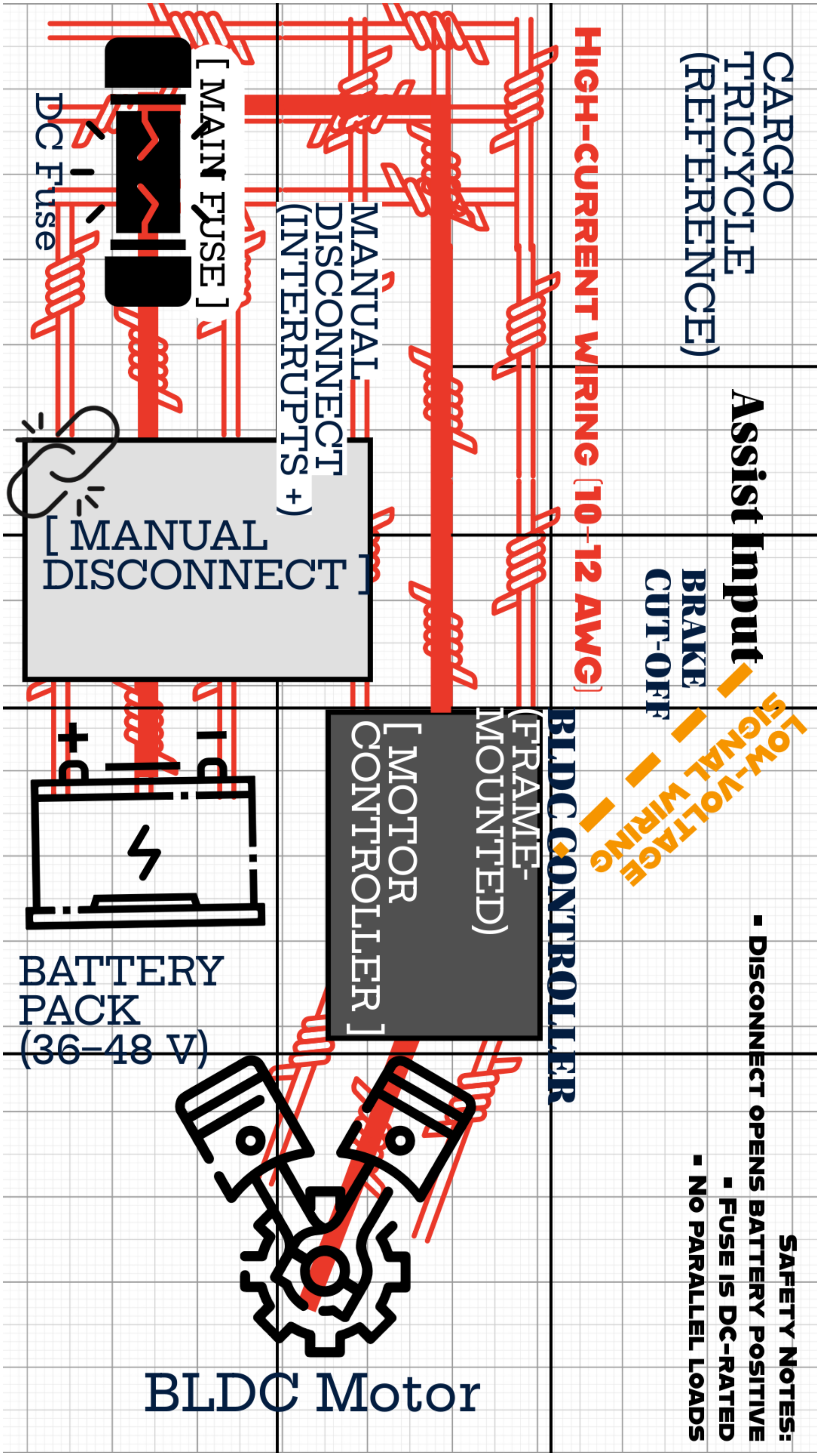
5.2 External Axle Torque Coupling -> safety-critical mechanical interface

How torque enters the axle without damage; fail-safe slip behavior.



5.3 Electrical Installation and Safety Routing -> real electrical build

Component placement, routing, disconnect + fuse placement.



6. Proposed Validation Methodology

6.1 Validation Objectives

The validation process evaluates whether the retrofit architecture satisfies its primary design claims:

1. Preservation of manual rideability
2. Safe and controllable assist torque delivery
3. Fail-safe behavior under electrical and mechanical faults
4. Compatibility with informal workshop installation practices

The methodology is structured to demonstrate functional safety and real-world operability rather than laboratory-only performance.

6.2 Test Vehicle Eligibility

Validation is conducted on cargo tricycles representative of the target deployment class. Eligible vehicles must meet the architectural assumptions defined in Section 3, including:

- Solid rear axle without differential
- Chain-driven drivetrain
- Steel frame construction
- Functional mechanical braking system
- Absence of axle damage or excessive play

Vehicles failing these criteria are excluded to maintain consistency with the defined operating envelope.

6.3 Installation Verification

Because installability is a core research claim, the retrofit is installed using only standard workshop tools (spanners, torque wrench, alignment tools, multimeter). No machining or frame modification is permitted.

Installation verification focuses on confirming:

- Frame-supported motor mass
- Absence of axial loads on axle bearings
- Proper alignment of the reduction stage
- Free back-driving of the axle and drivetrain

This stage validates compatibility with informal repair ecosystems.

6.4 Electrical Commissioning Tests

Electrical commissioning is performed with the rear wheels unloaded to isolate drivetrain behavior.

Initial power-on tests verify:

- Zero torque at zero input
- Smooth torque ramp with assist demand
- Immediate torque disable via brake cut-off
- Absence of unintended motor activation

These tests confirm correct implementation of the safety chain described in Section 4.3.

6.5 Controlled Ride Testing

Ride testing is conducted in three stages to progressively increase system load.

6.5.1 Low-Speed Functional Test

The vehicle is lightly loaded and operated at walking speed. Evaluation criteria include:

- Smooth torque onset
- Absence of drivetrain noise or binding
- Predictable assist disengagement

6.5.2 Nominal Load Test

The vehicle is loaded to typical operating mass and tested on flat terrain. Key observations:

- Rider control authority remains dominant
- No step changes in assist torque
- Stable interaction between human and electric power

6.5.3 High-Load / Incline Test

Testing under high load and incline conditions evaluates peak assist demand:

- Thermal stability of motor and controller
- Coupler slip behavior under overload
- Preservation of rider control in high-torque scenarios

6.6 Fail-Safe Behavior Testing

Intentional fault scenarios are introduced to validate safety claims.

Fault Condition	Expected Outcome
Battery removal	Immediate loss of assist
Fuse activation	Immediate loss of assist
Disconnect opening	Immediate loss of assist
Brake activation	Immediate torque cut-off
Mechanical transmission failure	Free pedaling maintained

Successful validation requires that **all faults result in zero assist without loss of manual rideability.**

6.7 Evaluation Criteria

The system is considered validated if the following criteria are met:

- Manual rideability indistinguishable from stock vehicle
- Smooth and predictable assist behavior
- Immediate brake override response
- Passive mechanical disengagement under overload
- Acceptable noise and vibration levels

Failure of any criterion requires system rework and retesting.

6.8 Deployment Readiness Indicators

Post-validation monitoring is recommended to observe real-world durability. Follow-up inspections focus on:

- Coupler wear
- Fastener torque retention
- Cable strain and routing integrity

These observations inform long-term reliability assessment

7. IMPACT AND DEPLOYMENT POTENTIAL

7.1 Deployment Context

The proposed retrofit targets human-powered cargo tricycles operating within informal transport economies. These vehicles form a critical logistics layer in many cities across South Asia, Africa, and Southeast Asia, where they are used for last-mile goods movement, waste collection, and small-scale commerce.

Operators in this segment face three persistent constraints:

- High physical workload and fatigue
- Low daily income margins
- Limited access to formal vehicle financing or replacement options

Full electrification through vehicle replacement is typically economically inaccessible. Consequently, retrofit solutions that preserve existing vehicles while augmenting rider capability represent a potentially scalable intervention.

The system described in this work is intentionally designed to operate within the economic, technical, and repair constraints of this environment.

7.2 Economic Feasibility of Prototype Implementation

A first-generation prototype was costed using components sourced within Indian urban supply ecosystems (Delhi, Bengaluru, Pune). The estimate reflects small-batch laboratory prototyping using commercially available components and limited custom fabrication.

The total cost of a single working prototype is estimated at:

₹30,000 – ₹35,000 per unit

The primary cost drivers are:

Subsystem	Approx. Cost (₹)	Share
Battery pack (LFP)	10,000	~30%
Motor + reduction stage	11,500	~35%
Controller + safety hardware	3,350	~10%
Mechanical fabrication & mounting	4,500	~14%
Wiring, controls, contingency	3,400	~11%

Energy storage dominates total cost, consistent with other light electric vehicle platforms.

This estimate reflects first-iteration engineering rather than optimized manufacturing.

7.3 Affordability Relative to Vehicle Replacement

In the target markets, replacing a cargo tricycle with a commercially available electric cargo vehicle typically costs ***₹90,000–₹140,000***.

The retrofit therefore represents approximately:

25–35% of the cost of full vehicle replacement.

This cost differential is central to the deployment strategy:

- Existing vehicle assets remain usable
- Operators avoid large capital expenditure
- Financing requirements are significantly reduced

The retrofit model aligns more closely with the incremental investment patterns typical of informal transport operators.

7.4 Pathways to Cost Reduction

The prototype cost reflects low-volume procurement and manual fabrication. Several cost reductions are expected under scaled production:

Battery procurement
Bulk cell purchasing and standardized pack design could reduce battery cost by 20–30%.

Reduction stage optimization
Transition from industrial planetary gearboxes to purpose-built cycloidal reducers may reduce cost while improving durability.

Coupler manufacturing
Batch machining or casting of the axle coupler could substantially reduce per-unit fabrication cost.

Bracket standardization
Vehicle-agnostic mounting kits could reduce custom fabrication labor.

A conservative projection suggests that scaled production could reduce total system cost by **20–30%**, bringing the expected unit cost to approximately:

₹22,000 – ₹28,000 per retrofit kit.

7.5 Compatibility with Informal Repair Ecosystems

A central deployment requirement is compatibility with small, independent workshops rather than specialized service centers.

The retrofit was designed to require:

- No frame cutting or welding
- No axle machining
- Only standard mechanical and electrical tools
- Modular, replaceable subsystems

This approach allows installation, maintenance, and repair to be integrated into existing repair networks. Such compatibility is critical for long-term sustainability and adoption.

7.6 Operational Impact for Riders

Electric torque assistance has the potential to influence daily operation in several ways:

- Reduced rider fatigue during high-load transport
- Improved ability to operate on inclines or poor road surfaces
- Potential increase in daily trip count or payload capacity
- Reduced physical strain over long working hours

These effects may translate into increased earning potential and improved occupational health, although longitudinal field studies are required to quantify these outcomes.

7.7 Scalability Considerations

The architecture is designed for scalability across multiple vehicle variants:

- Single-speed cargo tricycles
- Pedicabs and passenger rickshaws
- Utility trikes used for municipal services

The reliance on externally mounted, non-invasive subsystems enables adaptation without redesign of the host vehicle platform.

7.8 Summary

The prototype demonstrates that a mechanically non-invasive torque-assist retrofit can be implemented at a cost substantially below full vehicle electrification. Combined with compatibility with informal repair ecosystems, this suggests a plausible pathway toward scalable deployment in low-income transport sectors.

The next section discusses the current limitations of the prototype and directions for future research.

---

8. Limitations and Future Work

8.1 Mechanical Integration Constraints

The proposed retrofit architecture is designed to minimize invasive modification to existing vehicles. However, mechanical integration remains dependent on vehicle-specific geometry and available mounting clearances.

Key limitations include:

• Variation in rear axle geometry across vehicle models
• Limited mounting envelope for motor–reducer assembly
• Dependence on frame structural integrity in aging vehicles
• Need for vehicle-specific mounting bracket adaptation

The current prototype demonstrates feasibility on a representative vehicle platform, but a scalable deployment would require a library of modular mounting kits tailored to common vehicle classes (e.g., small hatchbacks, sedans, light commercial vehicles).

Future work will focus on:

• Parametric mounting bracket design
• Structural load testing across vehicle classes
• Finite Element Analysis (FEA) validation of frame stress distribution

---

8.2 Energy Storage and Thermal Considerations

The present work focuses on the drivetrain retrofit architecture rather than battery pack optimization. As a result:

- Battery thermal management is assumed to be passive
- Energy storage packaging constraints are not fully optimized
- Long-duration high-load operation has not been extensively validated

Thermal performance becomes critical in high ambient temperature regions and dense urban traffic conditions.

Future work will investigate:

- Passive and active battery thermal management strategies
- Pack placement optimization for safety and cooling
- Long-cycle durability testing under urban duty cycles

8.3 Control System Maturity

The control strategy implemented in the prototype prioritizes safety, simplicity, and deterministic behavior. While sufficient for validation, it represents a first-generation control architecture.

Current limitations:

- Rule-based torque assist strategy
- Limited vehicle state estimation
- No predictive energy management
- No learning-based optimization

Future development directions include:

- Model-based torque blending and drivability optimization
- Integration of vehicle state observers
- Predictive control using drive-cycle forecasting
- Data-driven efficiency optimization

8.4 Safety Certification and Regulatory Pathways

Although the system is designed with layered safety principles, formal certification remains a significant barrier to large-scale deployment.

Outstanding challenges:

- Absence of standardized certification frameworks for retrofit electrification
- Variability in regional homologation requirements
- Need for formalized failure-mode testing and validation

Future work will include:

- Formal Failure Modes and Effects Analysis (FMEA)
- Compliance mapping with emerging EV retrofit regulations
- Collaboration with regulatory bodies and certification agencies

---

8.5 Scalability and Manufacturing

The prototype demonstrates technical feasibility but does not yet represent a mass-manufacturable product.

Current constraints:

- Prototype-grade fabrication methods
- Limited supply chain validation
- Manual installation procedure

Future research will focus on:

- Design for manufacturability (DFM)
- Modular kit standardization
- Installation workflow optimization for technicians
- Cost reduction through volume manufacturing

8.6 Field Testing and Long-Term Reliability

The presented results are based on prototype-level validation and engineering analysis. Extended field trials are required to quantify:

- Long-term drivetrain wear characteristics
- Environmental durability (dust, water, vibration)
- Real-world energy savings and emissions reduction
- User adoption and operational reliability

Future work will include pilot deployments and real-world data collection programs.

---

9. Conclusion

This work presents a practical retrofit architecture for hybridizing existing internal combustion engine (ICE) vehicles through the addition of an external electric assist drivetrain. The proposed system is intentionally designed around real-world constraints of informal and cost-sensitive vehicle retrofitting, prioritizing non-invasive integration, safety, modularity, and scalability.

The paper contributes the following:

• A retrofit-first system architecture that avoids major modifications to the original vehicle drivetrain
• A mechanically independent electric propulsion path coupled via an external axle torque interface
• A layered electrical and safety architecture suitable for real-world installation conditions
• A prototype implementation demonstrating the physical feasibility of mounting, coupling, and wiring in an existing vehicle platform

By separating high-level system architecture from physical implementation, the work demonstrates both conceptual soundness and engineering realizability. The inclusion of real installation procedures and safety routing elevates the system from a theoretical design to a prototype-validated retrofit solution.

The broader motivation of this research is the electrification of the existing global vehicle fleet. Full vehicle replacement is capital-intensive and slow, particularly in regions where large numbers of legacy vehicles remain in operation. Retrofit hybridization represents a transitional pathway capable of delivering meaningful reductions in fuel consumption and emissions without requiring full vehicle replacement.

The presented prototype validates that:

• Non-invasive electrification of existing vehicles is technically feasible
• Safety-conscious installation can be achieved using modular subsystems
• The architecture can serve as a foundation for scalable retrofit kits

While further work is required in control refinement, certification, thermal management, and long-term validation, the system establishes a credible engineering pathway toward deployable retrofit electrification.

This research supports the thesis that incremental electrification of existing vehicles can play a meaningful role in accelerating the global transition toward lower-emission transportation systems.